

Matematica - Clasa a 6-a

Exercitii: Capitol IV - Numere rationale ($\mathbb{Q}$)

Nume: _____

Total: 100p + 10p oficiu

Data: _____

Ex. 1. Incercuieste ADEVARAT sau FALS:

(10p)

- Orice numar natural este si rational A / F
- $|-3/4| = 3/4$ A / F
- $3/4 > 5/6$ A / F
- $1/3 + 1/4 = 2/7$ A / F
- $3/4 \times 4/3 = 1$ A / F
- $(-1/2)^2 = -1/4$ A / F
- $2^{(-3)} = -8$ A / F
- $(2/3)^2 = 4/9$ A / F
- Orice numar rational poate fi scris ca fractie A / F
- $3/4 : 3/8 = 2$ A / F

Ex. 2. Modul si comparare:

(10p)

a) Calculeaza modulul:

$$|-5/8| = ___ \quad |-1,4| = ___ \quad |3/4| = ___ \quad |0| = ___$$

b) Pune semnul corect <, > sau =:

$$\begin{array}{llll} 3/4 ___ 5/6 & 2/3 ___ 3/4 & -1/2 ___ -1/3 & 3/5 ___ 0,6 \\ 5/8 ___ 7/12 & -3/4 ___ -5/6 & 0,75 ___ 3/4 & 2/7 ___ 3/10 \end{array}$$

c) Ordoneaza crescator: $5/6, 3/4, 7/12, 2/3, 11/12$

$$___ < ___ < ___ < ___ < ___$$

Ex. 3. Adunare si scadere cu fractii:

(10p)

Calculeaza si simplifica la forma ireductibila:

$$\begin{array}{llll} 1/3 + 1/4 = ___ & 5/6 - 1/4 = ___ & 3/8 + 5/12 = ___ & 7/10 - 3/5 = ___ \\ -3/4 + 1/2 = ___ & 2/3 + (-1/6) = ___ & 5/6 + 7/9 = ___ & 11/12 - 3/8 = ___ \\ 3/5 + 7/15 - 2/3 = ___ & 7/8 - 5/6 + 1/12 = ___ & & \end{array}$$

Exercitii: Capitol IV (continuare)

Ex. 4. Inmultire si impartire:

(10p)

Calculeaza (simplifica inainte unde poti):

$$2/3 \times 3/4 = ___$$

$$5/8 \times 4/15 = ___$$

$$7/12 \times 9/14 = ___$$

$$3/5 \times 25 = ___$$

$$3/4 \div 1/2 = ___$$

$$5/6 \div 5/3 = ___$$

$$7/8 \div 7 = ___$$

$$9 \div 3/4 = ___$$

$$4/5 \div 8/15 = ___$$

$$(-3/4) \times 8 = ___$$

$$(-2/3) \div (-4/9) = ___$$

Ex. 5. Puteri cu exponent intreg:

(10p)

a) Calculeaza:

$$(2/3)^2 = ___$$

$$(-1/2)^3 = ___$$

$$(3/4)^2 = ___$$

$$(-2/5)^2 = ___$$

$$(1/2)^4 = ___$$

$$(-1)^3 = ___$$

$$(2/3)^0 = ___$$

$$(-3/4)^2 = ___$$

b) Calculeaza puteri cu exponent negativ:

$$2^{-2} = ___$$

$$3^{-1} = ___$$

$$(1/2)^{-3} = ___$$

$$(2/3)^{-2} = ___$$

$$5^{-2} = ___$$

$$(3/4)^{-1} = ___$$

$$4^{-1} = ___$$

$$(1/3)^{-2} = ___$$

Ex. 6. Calculeaza respectand ordinea operatiilor:

(15p)

(Scrie rezolvarea pas cu pas!)

a) $1/2 \times (3/4 + 1/6) =$

b) $3/4 \div (1/2 - 1/3) + 5/6 =$

c) $(1/2)^2 - 3/4 \times 2/3 + 1/6 =$

d) $[2/3 + (1/4 - 1/6)] \times (-3) =$

e) $(-2/3)^2 + 1/2 \times (1/3 - 5/6) =$

Exercitii: Capitol IV (continuare)

Ex. 7. Rezolva ecuatiile in $\mathbb{Q}$:

(15p)

(Rezolva pas cu pas si verifica!)

a) $x + 3/4 = 5/4$

$x =$ _____

b) $x - 2/3 = 1/6$

$x =$ _____

c) $3x = 5/4$

$x =$ _____

d) $-2x = 3/8$

$x =$ _____

e) $2x + 1/3 = 5/6$

$x =$ _____

f) $3x - 1/2 = 5/4$

$x =$ _____

g) $x/3 - x/4 = 1/6$

$x =$ _____

h) $x/2 + x/3 = 5/6$

$x =$ _____

i) $3(x - 1/2) = x + 5/6$

$x =$ _____

j) $2(x + 1/4) = 3x - 1/2$

$x =$ _____

k) $4(x - 1/3) = 2(x + 1/6)$

$x =$ _____

l) $5x/6 - 1/2 = x/3 + 1/4$

$x =$ _____

Exercitii: Capitol IV (continuare)

Ex. 8. Probleme rezolvate cu ecuatii:

(15p)

a) Suma a doua numere este $7/4$. Unul este $5/8$. Gaseste celalalt.

Raspuns: ---

b) Un numar scazut din dublul sau da $5/6$. Care este numarul?

Raspuns: ---

c) Jumatatea unui numar plus o treime din el dau 10. Care e numarul?

Raspuns: ---

d) Un produs are pretul redus cu $1/4$ si ajunge la 45 lei. Cat costa initial?

Raspuns: --- lei

e) Intr-o clasa, $3/8$ din elevi sunt fete. Sunt 15 fete. Cati elevi total?

Raspuns: --- elevi

f) Maria a cheltuit $2/5$ din bani pe carti si $1/4$ pe mancare.

I-au ramas 21 lei. Cati bani avea initial?

Raspuns: --- lei

Exercitii: Capitol IV (continuare)

Ex. 9. Calcule rapide - scrie doar rezultatul:

(5p)

a) $(1/2 + 1/3) \times 6 = ______$

b) $(3/4)^2 - (1/4)^2 = ______$

c) $2^{-2} + (1/2)^2 = ______$

d) $(-1/3)^2 \times (-3)^2 = ______$

e) $|-3/4| + |1/4| - |-1| = ______$

Ex. 10. Gaseste necunoscuta:

(5p)

a) $______ + 3/4 = 5/4 \Rightarrow x = ______$

b) $______ \times (-2/3) = 4/3 \Rightarrow x = ______$

c) $3/4 \div ______ = 3/2 \Rightarrow x = ______$

d) $|______| = 5/8 \Rightarrow x = ______ \text{ sau } ______$

e) $x^2 = 9/16 \Rightarrow x = ______ \text{ sau } ______$

Ex. 11. Ordoneaza descrescator:

(5p)

a) $-1/2, 3/4, -2/3, 1/6, -5/12$

$______ > ______ > ______ > ______ > ______$

b) $0,6; 5/8; -0,7; 3/4; -7/10$

$______ > ______ > ______ > ______ > ______$

BONUS (10 puncte)

Ex. B1. Demonstrează ca expresia nu depinde de x :

(4p)

$$E = (x + 1/2) \times (x - 1/2) - x^2 + 1/3$$

(Indicație: extinde $(x+1/2)(x-1/2) = x^2 - 1/4$)

$$E = \text{---} \quad (\text{nu depinde de } x!)$$

Ex. B2. Ecuație cu fracții mai complexă:

(3p)

$$(x - 1/2)/3 + (x + 1/4)/2 = 7/6$$

$$x = \text{---}$$

Ex. B3. Inecuație în $\mathbb{Q}$:

(3p)

$$2x/3 - 1/2 \leq x/4 + 5/6$$

$$x \text{ --- } \text{---} \quad (\text{soluții: ...})$$

Barem de corectare

Ex. 1 10p 10 x 1p

Ex. 2 10p a) 4x1p b) 8x0.5p c) 2p

Ex. 3 10p 10 calcule x 1p

Ex. 4 10p 11 calcule x 1p - 1p

Ex. 5 10p a) 8x0.5p b) 8x0.5p+2p

Ex. 6 15p 5 expresii x 3p (calcul+raspuns)

Ex. 7 15p 12 ecuatii x 1p (rezolvare+verif.)

Ex. 8 15p 5 calcule x 3p